VIVO VPN 14:12 86%

Sonnet 4.5

Is magnetism symmetrical?

Is attraction and repulsion symmetrical?

If I take two identical rectangular neodymium magnets and I map them in the faces as “1” and “2”, and in the body “3”, “4”, “5” and “6”, and test all 64 possible combinations (i.e., switching “1” and “2” and rotating “3”, “4”, “5” and “6”), leaving one magnet free on table while I approach the other slowly with my fingers, what would be the outcome?

Reply to Claude

VIVO VPN 14:12 86%

If I take two identical rectangular neodymium magnets and I map them in the faces as “1” and “2”, and in the body “3”, “4”, “5” and “6”, and test all 64 possible combinations (i.e., switching “1” and “2” and rotating “3”, “4”, “5” and “6”), leaving one magnet free on table while I approach the other slowly with my fingers, what would be the outcome?

Why?

Where are the poles located?

Does rotating the body of the magnet (i.e., “3”, “4”, “5” and “6”) influence the outcome?

If we are at a repulsive combination we rotate one of magnets 180 degrees (i.e.,

Reply to Claude

VIVO VPN 14:12 86%

Does rotating the body of the magnet (i.e., “3”, “4”, “5” and “6”) influence the outcome?

If we are at a repulsive combination and we rotate one of magnets 180 degrees (i.e., switching the faces “1” and “2”), what will be the outcome — will it keep in repulsion or will it switch to attraction?

Looking at your image, I can see two small rectangular neodymium magnets on a table - this is a great hands-on way to explore magnetism!

Let me answer your questions systematically:

Is magnetism symmetrical? Is

Reply to Claude

VIVO VPN 14:12 86%

Let me answer your questions systematically:

Is magnetism symmetrical? Is attraction and repulsion symmetrical?

Yes, magnetism exhibits several symmetries. The force between two magnets is symmetric in that $F_{12} = -F_{12}$ (Newton's third law). Attraction and repulsion are also symmetric - if configuration A attracts configuration B, then swapping which magnet is which gives the same attraction.

The 64 combinations experiment

Your rectangular magnets are magnetized through their thick (faces "1" and "2" are the poles). Here's what you'll find:

Reply to Claude

VIVO VPN 14:12 86%

The 64 combinations experiment

Your rectangular magnets are magnetized through their thickness (faces "1" and "2" are the poles). Here's what you'll find:

You won't actually get 64 different outcomes - you'll get only 2:

- **Attraction** (opposite poles facing)
- **Repulsion** (same poles facing)

Why?

The poles are located on the **flat faces** (your "1" and "2"). One face is North, the other is South. The magnetization runs perpendicular to these faces, through the thickness of the magnet.

Does rotating the body ("3", "4",

Reply to Claude

Why?

The poles are located on the **flat faces** (your "1" and "2"). One face is North, the other is South. The magnetization runs perpendicular to these faces, through the thickness of the magnet.

Does rotating the body ("3", "4", "5", "6") influence the outcome?

No. Rotating a magnet around the axis perpendicular to the pole faces does nothing. The magnetic field has cylindrical symmetry around this axis. Whether you align the long edges or have them at 45° or 90° makes no difference to attraction/repulsion.

The only thing that matters is **which face**

Reply to Claude

VIVO VPN 14:13 86%

Does rotating the body ("3", "4", "5", "6") influence the outcome?

No. Rotating a magnet around the axis perpendicular to the pole faces does nothing. The magnetic field has cylindrical symmetry around this axis. Whether you align the long edges or have them at 45° or 90° makes no difference to attraction/repulsion.

The only thing that matters is **which face is pointing toward the other magnet.**

If we flip one magnet 180° (switching faces "1" and "2")?

It will switch from repulsion to attraction (or vice versa).

If you had:

Reply to Claude

VIVO VPN 14:13 86%

If we flip one magnet 180° (switching faces "1" and "2")?

It will switch from repulsion to attraction (or vice versa).

If you had:

- Magnet A: North facing up
- Magnet B: North facing down (repulsion)

After flipping B 180°:

- Magnet A: North facing up
- Magnet B: South facing down (attraction)

The magnetic field doesn't care about the body orientation - only which pole faces which direction!

Reply to Claude